

Exam 1 review day

Reminders and agenda

- + Exam 1 on Friday
 - + Bring a calculator!
 - + Review questions and solutions on the course website
- + Today:
 - + Quick discussion about outliers activity from Monday
 - + Exam 1 review

Outliers

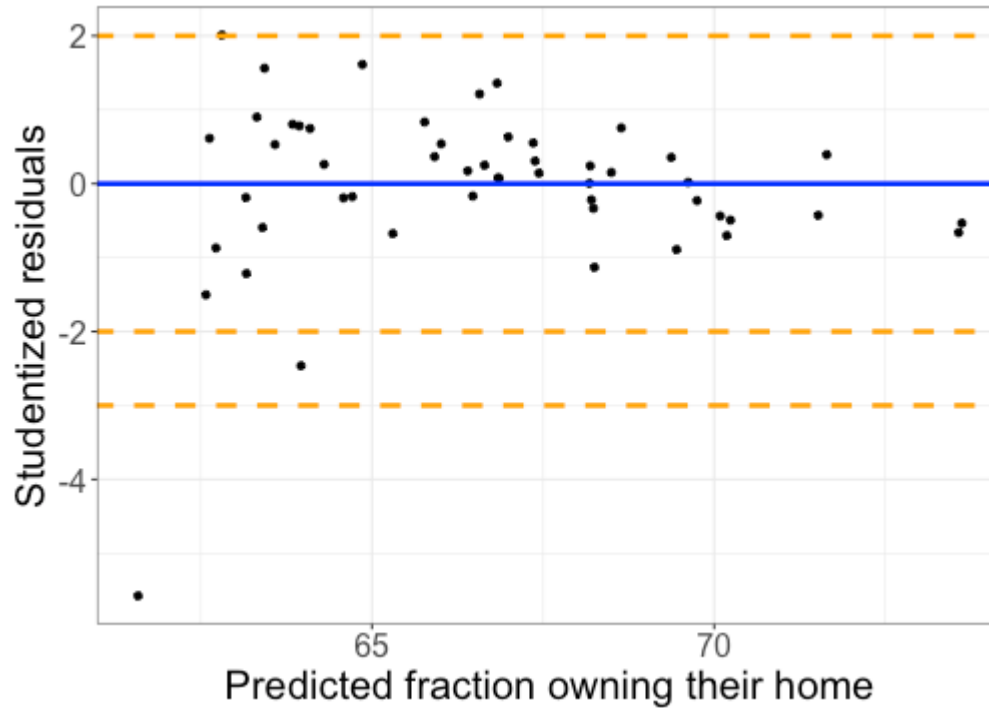
When can we *remove* potential outliers from the data?

- + When the observation is a clear error (e.g., a negative value for height)
- + When the observation comes from a different population than our population of interest (e.g., if we want to fit a model about dogs, then we can remove a cat from the data)

In all other situations:

- + Report the results (estimated coefficients, confidence intervals, test statistics, etc.) both **with** AND **without** the potential outlier(s)

Last time: exploring outliers



What point is the outlier here?

Last time: exploring outliers

DC: Percent urban -- 100%, Percent owning home -- 42%

Is this observation a clear *error*?

Last time: exploring outliers

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No -- these are believable numbers, given what we know about DC

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Does DC belong to a different population than our population of interest?

Last time: exploring outliers

DC: Percent urban -- 100%, Percent owning home -- 42%

Is this observation a clear *error*?

No -- these are believable numbers, given what we know about DC

Does DC belong to a different population than our population of interest?

Maybe -- I think you can make an argument either way

Last time: exploring outliers

How should we handle DC?

- + If we think DC does not belong in our population of interest, could remove
- + Otherwise, report results with and without DC

RMS E (root mean squared error)

squared error: $\sum_{i=1}^n (y_i - \hat{y}_i)^2$ (SSE)

mean squared error: $\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \frac{SSE}{n}$

root mean squared error: $\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} = \sqrt{\frac{1}{n} SSE}$

RMS E: how far our predictions tend to be from the observed responses

SSE depends on:

- relationship between explanatory + response
- scale of response variable
- # of observations

RMS E depends on:

- relationship between explanatory + response
- scale of response variable

Test Statistic:

$$H_0: \beta_1 = b$$

$$(e.g., H_0: \beta_1 = 0)$$

estimate
from model

$$t = \frac{\hat{\beta}_1 - b}{SE\hat{\beta}_1}$$

← value under H_0

← standard error

= distance between an estimated slope and the value hypothesized under H_0 , in standard deviations

(close 2 of forecasting β_0, β_1)

under H_0 : $t \sim t_{n-2}$

$$t \text{ w/large df} \approx N(0,1)$$

⇒ e.g., 3 standard deviations is large



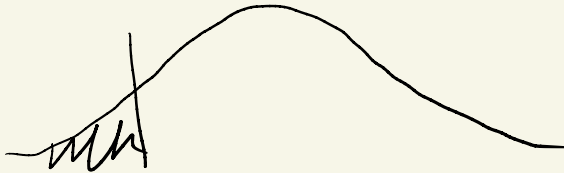
area corresponding to a p-value;

Two-tailed test, e.g. $H_A: \beta_1 \neq 0$

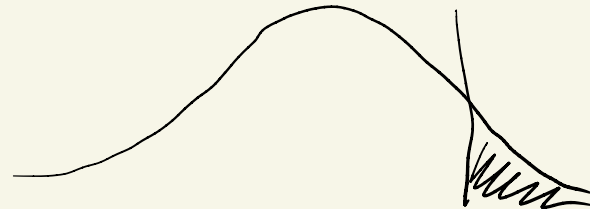


One-tailed test:

$H_A: \beta_1 < 0$



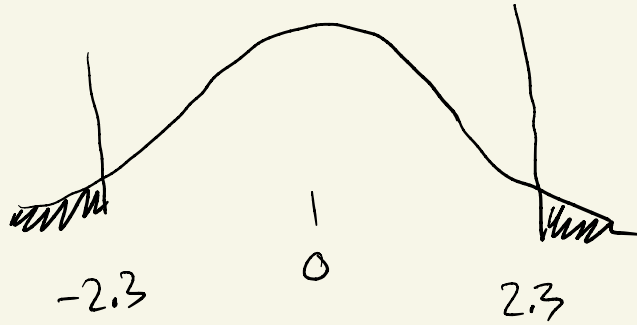
$H_A: \beta_1 > 0$



e.g.

$$t = -2.3$$

$$H_A: \beta_1 \neq 0$$



p-values < 0.01

quite unusual

p-values > 0.1

not unusual

Residual: $y_i - \hat{y}_i$

(how far away is our prediction from the truth, in original scale of response variable)

Studentized residual: $\frac{y_i - \hat{y}_i}{\hat{\sigma}_\varepsilon}$

← residual

← standard deviation of the residuals, ignoring i th observation

Transformations

$$y = \beta_0 + \beta_1 x + \varepsilon$$

β_1 : average change in y associated with a 1-unit increase in x

$$y = \beta_0 + \beta_1 \sqrt{x} + \varepsilon$$

β_1 : average change in y associated with a 1-unit increase in $\sqrt{x}$

$$y = \beta_0 + \beta_1 \log(x) + \varepsilon$$

β_1 : a 1% increase in x is associated with a change in y of $\beta_1/100$, on average

linear:

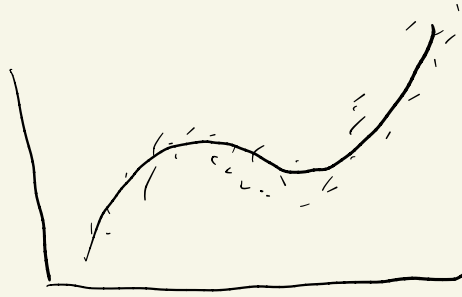


Polynomial order 2:



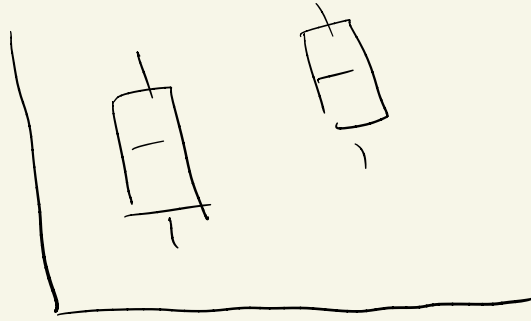
change in
direction

Polynomial order 3:



← 2 changes
in direction

boxplots



- compare different groups
- describe center (median)
spread (IQR) of
distribution